

Transcribed Notes: Pinhole Model

Pinhole model

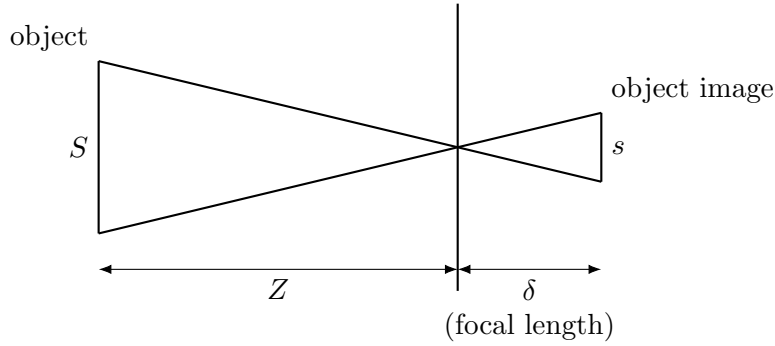


Figure 1: Pinhole camera geometry used in the notes.

By similar triangles, we have

$$\frac{Z}{S} = \frac{\delta}{s}.$$

Hence,

$$Z = \frac{\delta S}{s}.$$

If δ is fixed and the true object size S is known, define

$$k = \delta S.$$

Then

$$Z = \frac{k}{s}.$$

Using this model, for each type of object, we can calibrate (find k) using an image where the actual distance Z to the object is known.

Case of containers and spreaders

Container and spreader lengths and widths can be said to be the same. Assume that we know the exact values for these beforehand.

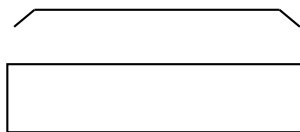


Figure 2: Simple spreader/container sketch from the notes.

Then

$$Z_s = \frac{\delta S_s}{s_s}, \quad Z_c = \frac{\delta S_c}{s_c}.$$

But since $S_s = S_c$ (call this common length ℓ), we can find δ using a single calibration image for the spreader:

$$\delta = \frac{Z_s s_s}{\ell}.$$

Then the same value can be used for both the container and the spreader, and

$$k = \delta \ell.$$

When we have more than one calibration point

If we have more than one calibration point, we can fit a line in terms of $1/s$:

$$Z = m \left(\frac{1}{s} \right) + c.$$

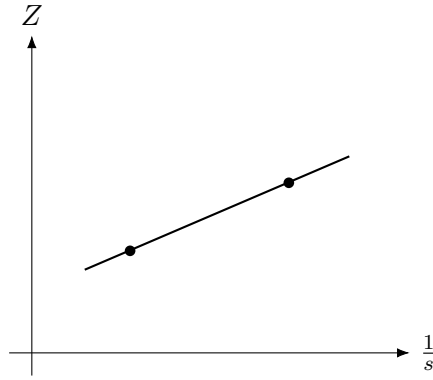


Figure 3: Linear calibration relationship sketched in the notes.

Again, since container and spreader lengths and widths are the same, we can use the same m and c to both of them.